Himalayan Glacial Disaster — DRR Concept Diagram

Support options drawing on Japanese technology, following the 25–26 August 2026 Bhote Koshi–Trishuli flood in Nepal (working draft)

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**What happened at the source is now largely understood.
Downstream there were 6–10 hours. They did not turn into evacuation.**

1 What happened

08:37	**The north face of Langtang Lirung fails, bedrock and all** (0.2 km² of rock and ice, 1,200 m drop). It **briefly dams the Lhende**, then breaches as one surge. M5.2 / Mw 5.7 **came from the collapse**
08:50 / 09:20	Stage gauges **stop transmitting** (= destroyed). 09:05 DHM becomes aware, **16 min after the wave crossed the border**
09:15	Mass SMS alert: **679,295 messages**. Only 4 districts
11:45	**A second collapse on the same slope** (M4.2). Search teams are entering the valley — **no warning issued**
15:20 / 16:00	Reaches Devghat → peak **6.57 m** (no second peak)

About 10 hours from collapse to the lowest reach · 1,114 dead nationwide, ~4,500 missing

This chronology took a while to settle. First USGS report → IRDR (previous-day collapse + long impoundment) → satellite analysis (no dam), before settling on **ICIMOD / USGS: a brief blockage and breach.** With no instruments upstream, everything rested on remote inference. A GLOF struck the same corridor in July 2025 (11 dead).

2 Diagnosis — most bodies were found where there had been time to escape

District (upstream → downstream)	Margin after alert	Bodies recovered
Rasuwa	−25 min	27
Nuwakot	+5 min	95
Dhading	~1.5–2 h	55
Gorkha	~3–4 h	65
Tanahun	~4–5 h	38
Chitwan	6 h 05 min	321
Nawalparasi E (not alerted)	~7–8 h	216
Nawalparasi W (not alerted)	~8–9 h	170

A decisive caveat: "bodies recovered" is **where a body was found, not where the person died**, and **recovery is rate-limited by regaining access** (from 29 Aug to 1 Sep the upstream counts caught up: Rasuwa 13 → 27, Nuwakot 51 → 95). That is exactly why **the verification survey (A-0) must precede any equipment donation**. The shape still holds: 489 people — **49.5% of this corridor** — were recovered in the four districts that were never alerted; the three lowest districts alone hold **71.6%**; Rasuwa, closest to the source, holds **2.7%**.

3 Four gaps

GAP 1 What happened at the source is not knowable quickly
USGS → IRDR (prolonged impoundment) → satellite (no dam), settling on **ICIMOD / USGS: a brief blockage and breach**. Readings diverge because there are no upstream instruments and everything is remote inference. That, in itself, is the case for putting monitoring there.

GAP 2 Monitoring watches only for a rise in water
The Rasuwagadhi gauge (446.22) showed no prior change. The upstream stations **stopped transmitting together after 08:40**, and that silence — itself the evidence of destruction — was logged as "missing data". **The renewed rise on 30 Aug was likewise noticed by the river's roar, not by an instrument.** Neither a falling stage nor missing data is treated as an anomaly.

GAP 3 "A flood has occurred" does not move people
670,000 messages went out at 09:15, yet the wave reached Chitwan **six hours later**. What was needed was not notice that something had happened but "what time it arrives at your village", followed by tracking updates.

GAP 4 Casualties concentrated among those SMS cannot reach
Across 12 damaged hydropower projects **about 900 are unaccounted for, some 500 of them inside tunnels** (Upper Trishuli-1 alone: 576 missing, ~300 believed inside). 583 foreign nationals are also missing. Tunnels have neither signal nor escape routes; trekkers have neither Nepali-language SMS nor community networks. **Neither group is in the design scope of the existing system.**

4 The measures as a whole — laid out along the lead-time axis

AT THE SOURCE

B-4 / TRANSBOUNDARY — Peacetime monitoring
Slopes >30°, above 3,000 m, permafrost, near a large river
▶ ALOS-2/4 · Sentinel Asia

B-1 / DETECT — Collapse detection
Detects a collapse and alerts automatically, without waiting
▶ NIED Hi-net · microbarometers

B-3 / PREDICT — Blockage assessment
The channel was blocked twice here; both needed assessing
▶ Japan's emergency-survey rules

SOURCE MECHANISM — THE READING MOVED, BUT HAS SETTLED

USGS: the 08:37 collapse is the origin → IRDR: a 13:00 collapse the day before, breach after 18–19 h → satellite: no dam
Now: bedrock fails with the ice at Langtang Lirung, briefly dams the Lhende, then breaches as one surge

Orders of magnitude — only the excess constrains the source (now measured)

Frictional melt (even spending all of the 1,200 m drop — 11.8 kJ/kg — caps melt at 3.5% of the ice mass)	10^5–10^6 m³
18–19 h of channel blockage and impoundment (a catchment the size of the Lhende)	10^6 m³
Total volume past Devghat (includes baseflow; cannot constrain the source). Peak flow 5,850 m³/s	10^8 m³
↳ **the excess above baseflow — larger than the fallen mass (0.5–10×10^6 m³); most of the water was picked up en route**	2×10^7 m³

The renewed rise on 30 Aug — the same gap, reproduced four days later — the barrier lake was largely empty by 30 Aug
Flow rose again and NDRRMA alerted. Noticed not by an instrument but by **a report that the river's roar had grown**. Search suspended.

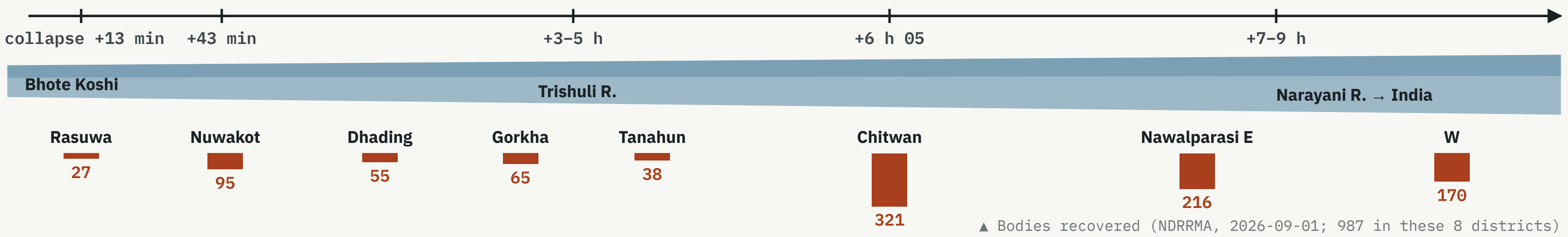
AFTER THE COLLAPSE — 100 KM IN UP TO 10 HOURS (THE ALERT WENT OUT; EVACUATION DID NOT FOLLOW)

B-2 / DISSEMINATE — Automatic sirens
Wired straight through — no human decision in the loop
▶ Tokachidake wire / vibration sensors

A-2 / DETECT — Track the wave with gauges
Trigger on rate of rise; missing data = maximum alert
▶ Crisis-management stage gauges, 6 → 30

A-1 / PREDICT — TOP PRIORITY — Arrival times, alerts by village
"Arrives 11:40. Reach high ground by 10:30" — then keep updating
▶ Flood Risk Line · timeline-based DRR

A-3 / DISSEMINATE — Alert extent = hazard extent
Defined by the river corridor, not by administrative district
▶ CAP-compliant · the L-Alert design



BASIN-WIDE, PEACETIME — **A-4** Protocols for tunnel workers and foreign nationals (cell broadcast / tunnel safety standards) **C-1** High-ground refuges and escape routes for every settlement **C-3** Hydropower plants as EWS nodes (monitoring obligations secured through loan and insurance conditions) **A-0** Verification survey (settle the blockage duration and water budget; census of where and when each death occurred and which alert was received)

Left: what happened at the source, as understood on 2026-09-03. Right: propagation of the flood wave after the collapse (the horizontal axis is not to scale). How much margin there was at the source depends on the reading, but **the 6–10 hours downstream were there on any of them**. Only three arrival times are measured — Syabrubesi 08:50, Betrawati 09:20, Devghat 15:20; the rest are interpolated. On the Trishuli, a 9 m rise at Galchchi and 7 m at Malekhu were recorded within 30 minutes. Bodies recovered is not a count by place of death, and recovery is rate-limited by regaining access.

5 Japanese technology and the bodies that would deliver it

DETECT — NIED / Japan Meteorological Agency
Hi-net, V-net; automatic seismic and infrasound detection of lahars
Stations sit inside Nepal, so the system **can be built without waiting for agreement with China**. It needs no identification of mechanism, so it works while views still diverge.

PREDICT — PWRI / ICHARM / MLIT
Flood Risk Line; timeline-based DRR
This disaster itself **left measured data with which to calibrate the model**.

CROSS-BORDER — JAXA / Sentinel Asia / ADRC
ALOS-2 and -4 (L-band SAR); International Disasters Charter
The source was **inside Nepal**, so China's warning system did not cover it. There is distinct value in **Japan observing and distributing neutrally, as a third country**.

DETECT — MLIT / instrument manufacturers
Crisis-management stage gauges; wire and vibration sensors
The "under ¥1 M, five years unpowered, retrofittable" specification adopted after Typhoon Lionrock in 2016 (24 dead in Iwaizumi).

DELIVER — MIC / mobile network operators
Emergency cell broadcast (ETWS / CB); L-Alert
Broadcasts to every handset in several languages without any subscriber list, and **reaches roaming foreign SIMs**.

STRUCTURAL — MLIT Sabo Dept. / JICA / JBIC and NEXI
Emergency survey of landslide dams; unmanned construction
A statutory procedure that runs impounded volume → breach discharge → inundation estimate within 24–48 h. Needed both for the brief initial blockage and for the barrier lake that overtopped on 28 Aug.

6 Sequence of implementation

WHEN	WHAT	WHO
0–3 mo	**A-0 verification survey** — settle the blockage duration and the water budget; census of place and time of death and of which alert was received. In parallel, satellite monitoring of the unstable mass	Joint academic survey team / JAXA / ADRC
0–1 yr	A-1 arrival-time forecasting; A-3 review of alert coverage; A-4 tunnel and foreign-national protocols	ICHARM / JICA / MIC
1–2 yr	B-1 source detection network (detects an upstream collapse and alerts automatically); B-2 automatic sirens; B-3 transfer of the channel-blockage procedure. Stage gauges to 30+	NIED / MLIT Sabo Dept.
2–5 yr	C-1 high-ground refuges and escape routes; C-2 land-use guidance; C-3 power plants as nodes, with financial conditions	JICA / JBIC and NEXI
ongoing	Build the transboundary data-sharing framework; institutionalise Japan's observation supply as a neutral third party	MOFA / ICIMOD / ADRC

On equipment dying within three years: require no mains power (five-year battery); use no consumables; duplicate the link (satellite IoT); hold spares and trained repairers locally; and pay against **95%+ annual uptime** rather than against delivery.

7 Outside the scope

Catching this collapse in advance from orbit is hard
Rock and ice breaking at once leaves little movement to see beforehand; the ground-temperature anomaly reported two days earlier was found afterwards. Satellites are good at three things: **keeping the lake inventory current, reading the situation after an event, and narrowing down which slopes to watch.**

This is beyond what check dams can hold
Catching a debris flow that runs more than 100 km with a structure is not realistic. What works is high-ground refuges, training dykes on tributaries, and designing rebuilt structures so they are not washed away.

Rainfall data could not see this one
No rain was involved in this flood. It seems better to **keep separate detection paths for rain-driven and glacier-driven floods, and share only how the warning reaches people downstream.**

DC Report cannot be the primary path in the mountains
Nepal sits near the western edge of the service area, and in deep valleys there will be places where the satellite cannot be seen high enough. **Start by measuring, on the ground, where reception actually works.**

Legend (what the colours mean)
- Detection & observation
- Prediction & decision
- Dissemination & alerting
- Action & structures
- Cross-border & institutions
- Human losses

Group A = immediate (within 1 yr) / Group B = medium term (2 yr; where Japan is distinctively strong) / Group C = structures and land use (3–10 yr). Priority = lead time created ÷ cost × capacity to maintain.

Glossary
- **DHM**: Dept. of Hydrology and Meteorology, Nepal
- **NDRRMA**: National Disaster Risk Reduction and Management Authority
- **GLOF**: glacial lake outburst flood
- **ICIMOD**: Int'l Centre for Integrated Mountain Development
- **ICHARM**: Int'l Centre for Water Hazard and Risk Management
- **CAP**: Common Alerting Protocol
- **ETWS / CB**: cell-broadcast emergency alerting
- **SAR**: synthetic aperture radar (sees through cloud)
- **Unmanned construction**: earthworks by remotely operated machines
- **Timeline DRR**: planning backwards from the arrival time

Find it early upstream. Deliver time downstream.

What this disaster lacked was not a means of knowing the danger, but a path from knowing to escaping. On 30 Aug, too, it was a human ear that noticed. Sources: GFZ, ICIMOD, Copernicus / Planet satellite analysis, IRDR, USGS, NDRRMA, MLIT, NIED, JAXA, JICA and others